PRACTICE EXAM 2 · EXAM STRATEGY - ALL DOMAINS

Practice exam 2

Twenty more questions, none repeating exam 1. Same rules, same clock.

TYPE Practice exam QUESTIONS 20 TOPIC All sections

admin@sthithapragna.com

> A fraud detection workload must flag transactions and publish to a Kafka topic with end-to-end latency under 500 milliseconds. The platform team's standard is that all new streaming work is built as declarative pipelines.
>
> Events arrive from Kafka.

What should the team do?

A Build it as a pipeline in continuous mode with the shortest available trigger.

B Build it as Structured Streaming in real-time mode on dedicated compute, since pipelines cannot run real-time mode.

C Build it as a pipeline on serverless, which supports real-time mode.

D Build it as a pipeline with a foreach batch sink writing to Kafka.

◆ PAUSE AND DECIDE

What should the team do?

A Build it as a pipeline in continuous mode with the shortest available trigger.

Continuous mode still runs micro-batches. It does not reach sub-second latency.

B Build it as Structured Streaming in real-time mode on dedicated compute, since pipelines cannot run real-time mode.

Real-time mode is unsupported on pipelines, classic or serverless, and Kafka to Kafka is what it supports.

C Build it as a pipeline on serverless, which supports real-time mode.

Factually false, and it is the option that sounds like a compromise between the requirement and the standard.

D Build it as a pipeline with a foreach batch sink writing to Kafka.

Solves the sink and ignores the latency.

ON THE EXAM

A team standard is not a requirement. When a stated latency target is unreachable by the standard approach, the target wins.

> A serverless job has three tasks. Two are Python wheel tasks and one is a notebook task. An engineer defines an environment with the required libraries and attaches it to all three by environment key.
>
> The notebook task fails, reporting the libraries are unavailable.

Why?

A. Notebook tasks cannot take a task environment specification. Their dependencies come from the notebook's environment panel.

B. An environment can be attached to only one task.

C. The environment version must match the notebook's runtime version.

D. Wheel tasks and notebook tasks cannot share a serverless job.

◆ PAUSE AND DECIDE

Why?

A Notebook tasks cannot take a task environment specification. Their dependencies come from the notebook's environment panel.

The documented exception. It is also why the high memory setting lives in that same panel.

B An environment can be attached to only one task.

Sharing one environment across tasks is the point of the environment key.

C The environment version must match the notebook's runtime version.

Invented requirement.

D Wheel tasks and notebook tasks cannot share a serverless job.

They can. Mixed task types in one job are normal.

ON THE EXAM

The two wheel tasks working and the notebook task failing is the tell. One task type behaves differently.

> A Structured Streaming job ingests from cloud storage. The business confirms the downstream dashboard is checked twice daily and a few hours of latency is acceptable. The stream currently runs continuously on an autoscaling cluster.
>
> The finance team has queried the cost.

What is the strongest change?

A Reduce the autoscaling minimum so the cluster is smaller between batches.

B Switch to the availableNow trigger on a schedule, so compute runs only while there is work.

C Switch to the once trigger, which processes a single micro-batch per run.

D Move the stream to a declarative pipeline in continuous mode.

◆ PAUSE AND DECIDE

What is the strongest change?

A Reduce the autoscaling minimum so the cluster is smaller between batches.

Autoscaling has known scale-down limitations for streaming, and the cluster still runs all day.

B Switch to the availableNow trigger on a schedule, so compute runs only while there is work.

Consumes everything available and stops. Same code, same checkpoint, a fraction of the compute.

C Switch to the once trigger, which processes a single micro-batch per run.

Deprecated from DBR 11.3 LTS, and a single micro-batch may not clear the backlog.

D Move the stream to a declarative pipeline in continuous mode.

Continuous mode keeps compute running, which is the cost being questioned.

ON THE EXAM

The business requirement is the lever. A continuous source does not compel continuous compute.

A serverless task sends onboarding emails to new customers through a third-party provider. During a transient network failure the task failed, and afterwards several customers reported receiving the email twice.

The task has no retry policy configured, and the code sends each email exactly once per invocation.

What should be changed?

A Set maximum retries to zero on the task.

B Disable serverless auto-optimization for this task, since it retries failed tasks by default.

C Move the task to a continuous job, which handles retries more carefully.

D Nothing; the provider duplicated the sends.

◆ PAUSE AND DECIDE

What should be changed?

A Set maximum retries to zero on the task.

Does not disable auto-optimization, which is what performed the retry.

B Disable serverless auto-optimization for this task, since it retries failed tasks by default.

Auto-optimization retries failed tasks by default. Sending email is not idempotent, so this needs at-most-once.

C Move the task to a continuous job, which handles retries more carefully.

Continuous mode retries automatically with backoff and cannot be configured.

D Nothing; the provider duplicated the sends.

Blames a third party for documented default behaviour on this platform.

ON THE EXAM

No retry policy is configured, and a retry still happened. That is the whole question.

> A team reads a Kafka topic carrying Avro payloads into a bronze streaming table. They select the value column and apply an Avro deserialisation function to it, and the job fails with a type error.
>
> A schema registry is configured and reachable.

What is missing?

A. The value must be cast to string before deserialisation.

B. Nothing is missing in the reader; the value column is binary, which is what from_avro expects.

C. The topic must be subscribed with the assign option rather than subscribe.

D. Avro payloads require the schema to be supplied as a table property.

◆ PAUSE AND DECIDE

What is missing?

A The value must be cast to string before deserialisation.

Casting binary Avro to string corrupts it. That is the fix for JSON, not for Avro.

B Nothing is missing in the reader; the value column is binary, which is what from_avro expects.

Kafka returns binary, and the Avro function consumes binary. The error is elsewhere, most likely the schema argument.

C The topic must be subscribed with the assign option rather than subscribe.

Any of subscribe, subscribePattern or assign works. It is not a type problem.

D Avro payloads require the schema to be supplied as a table property.

Invented mechanism.

ON THE EXAM

This inverts exam 1's habit. Kafka returning bytes is right for Avro, and the JSON fix would break it.

> A monthly reconciliation job runs an Auto Loader stream against a bucket holding several million files, using file notification with file events enabled. Each monthly run now takes hours before processing begins, though very little new data arrives.
>
> Correctness has never been in question.

What explains it?

A The file events cache expires; streams should run at least every seven days, so a monthly run falls back to a full directory listing.

B The checkpoint has grown too large to load.

C Backfill interval must be set manually for streams running less than daily.

D File events are unsupported for buckets above a certain file count.

◆ PAUSE AND DECIDE

What explains it?

A The file events cache expires; streams should run at least every seven days, so a monthly run falls back to a full directory listing.

A month is well outside the seven-day window, so the stored read position is invalid and a full listing is required.

B The checkpoint has grown too large to load.

Checkpoint size tracks files processed, and very little new data arrives.

C Backfill interval must be set manually for streams running less than daily.

Backfill is handled automatically once file events are enabled.

D File events are unsupported for buckets above a certain file count.

No such limit. The limit is on time between runs.

ON THE EXAM

Correctness has never been in question is the clue. The fallback is complete and correct, and slow.

> A daily job filters a large orders table to those whose product appears in a small recalled-products table, using an inner join on product id. Finance reports the recalled order totals are overstated.
>
> The recalled-products table lists some products twice, once per affected batch.

What is the correct fix?

A. Deduplicate the recalled-products table before the join.

B. Replace the inner join with a LEFT SEMI join, which returns each qualifying order exactly once.

C. Add SELECT DISTINCT to the join output.

D. Replace the inner join with a LEFT ANTI join.

◆ PAUSE AND DECIDE

What is the correct fix?

A Deduplicate the recalled-products table before the join.

Fixes today's data. The duplicates are legitimate, being one row per affected batch, so they will return.

B Replace the inner join with a LEFT SEMI join, which returns each qualifying order exactly once.

A semi join is a filter that cannot multiply rows, whatever the right side contains.

C Add SELECT DISTINCT to the join output.

Masks the duplication at the cost of a shuffle, and would collapse genuinely identical orders.

D Replace the inner join with a LEFT ANTI join.

Returns orders with no recalled product, which is the opposite set.

ON THE EXAM

The duplicates are legitimate here, which rules out deduplication as a fix. The join type is the thing that is wrong.

A pipeline defines an expectation on a silver table checking that a delivery date is not in the future. Metrics show violations being counted every day, and a downstream report still contains future-dated deliveries.

The expectation was written with no violation clause.

What is happening?

A The expectation is malformed and is being ignored.

B Warn is the default action, so violating records are written to the target and counted.

C The metrics are counting rows that were dropped, and the report reads a stale version.

D Expectations do not apply to date comparisons.

◆ PAUSE AND DECIDE

What is happening?

A The expectation is malformed and is being ignored.

The metrics prove it is being evaluated.

B Warn is the default action, so violating records are written to the target and counted.

An expectation with no violation clause counts rather than prevents. It is working exactly as written.

C The metrics are counting rows that were dropped, and the report reads a stale version.

Dropped rows would not appear downstream at all.

D Expectations do not apply to date comparisons.

Any SQL boolean expression works.

ON THE EXAM

Both observations are true at once. It is evaluated, and it lets the record through, because that is what warn does.

> A team federates a Snowflake source for exploratory analysis. It works well. The business now wants that data feeding an hourly production pipeline over several hundred million rows, with tight freshness requirements.
>
> The source is supported by both Lakehouse Federation and Lakeflow Connect.

What does Databricks recommend?

A Keep federation and scale up the SQL warehouse.

B Use Lakeflow Connect, which Databricks recommends when performance at volume or lower latency is a priority.

C Keep federation and enable result caching to reduce repeated load.

D Switch to catalog federation, which is more performance-optimized.

◆ PAUSE AND DECIDE

What does Databricks recommend?

A Keep federation and scale up the SQL warehouse.

The constraint is the source and the network, not Databricks compute.

B Use Lakeflow Connect, which Databricks recommends when performance at volume or lower latency is a priority.

Stated directly: where a source supports both, prefer Lakeflow Connect at volume or for lower latency.

C Keep federation and enable result caching to reduce repeated load.

Databricks query caching is not supported for federated queries at all.

D Switch to catalog federation, which is more performance-optimized.

True in general, and it is a different query path for catalog platforms rather than a faster setting.

ON THE EXAM

Federation is scoped to ad hoc reporting, BI and proof of concept. Growing past that scope is the signal to ingest.

> A finance analyst reconciling Databricks costs finds their monthly total is consistently higher than the invoice. They built the report from the billable usage system table, summing usage quantity grouped by month.
>
> To be careful, they excluded records with a negative usage quantity as obvious data errors.

What is wrong?

A. Negative quantities are retractions that cancel incorrect original records. Excluding them leaves the uncorrected values in.

B. The usage table reports list prices rather than negotiated ones.

C. They must also filter to record type RESTATEMENT to get corrected values.

D. Monthly grouping double counts usage spanning a month boundary.

◆ PAUSE AND DECIDE

What is wrong?

A Negative quantities are retractions that cancel incorrect original records. Excluding them leaves the uncorrected values in.

A correction is a negative retraction plus a restatement. Removing the negatives keeps the wrong figures.

B The usage table reports list prices rather than negotiated ones.

The table records quantities, and a pricing difference would be a consistent ratio.

C They must also filter to record type RESTATEMENT to get corrected values.

Filtering to restatements alone discards all correct original records.

D Monthly grouping double counts usage spanning a month boundary.

Invented.

ON THE EXAM

Exam 1 had this trap as a filter on record type. Here it is a filter on the sign of the quantity, which is more natural and equally wrong.

> A declarative pipeline ingesting from Kinesis must alert when it falls more than fifteen minutes behind. An engineer adds a streaming backlog notification with a backlog seconds threshold to the job that runs the pipeline task.
>
> During a confirmed three-hour backlog, no notification fires and no error appears anywhere.

What is the explanation and the alternative?

A Backlog alerting uses a ten-minute average, which smoothed the backlog away.

B Streaming backlog notifications are not supported for pipeline tasks. Use the event log's flow progress backlog metrics.

C Backlog seconds is not a supported metric; only bytes and records are.

D The threshold belongs on the pipeline rather than in the job's health rules.

◆ PAUSE AND DECIDE

What is the explanation and the alternative?

A Backlog alerting uses a ten-minute average, which smoothed the backlog away.

The averaging smooths brief spikes. A sustained three-hour backlog clears any average.

B Streaming backlog notifications are not supported for pipeline tasks. Use the event log's flow progress backlog metrics.

A documented limitation, and it fails silently. The event log carries backlog metrics in flow progress events.

C Backlog seconds is not a supported metric; only bytes and records are.

All four exist: bytes, records, seconds and files.

D The threshold belongs on the pipeline rather than in the job's health rules.

Health rules are the right place. The task type is the problem.

ON THE EXAM

Everything configured was correct. The unsupported combination fails silently, which is what makes it hard to diagnose.

> Two managed tables in the same schema receive identical write patterns. One is Z-ordered on two columns from an earlier project; the other has no clustering. Over six months the Z-ordered table's query performance degrades while the other's holds steady.
>
> Predictive optimization is enabled and no manual maintenance runs.

What explains it?

A Z-ordering degrades faster than no clustering and needs periodic re-running.

B Predictive optimization does not run ZORDER and ignores Z-ordered files, so that table receives less automatic maintenance.

C The unclustered table benefits from data skipping that Z-order disables.

D Predictive optimization only maintains tables with clustering keys defined.

◆ PAUSE AND DECIDE

What explains it?

A Z-ordering degrades faster than no clustering and needs periodic re-running.

Layouts do drift, and that does not explain one table getting less maintenance than its neighbour.

B Predictive optimization does not run ZORDER and ignores Z-ordered files, so that table receives less automatic maintenance.

A tuned table receives less automatic benefit than an untuned one, which is counterintuitive and documented.

C The unclustered table benefits from data skipping that Z-order disables.

Z-order does not disable data skipping. It is a layout technique that supports it.

D Predictive optimization only maintains tables with clustering keys defined.

It maintains managed tables generally, which is why the unclustered one is fine.

ON THE EXAM

The engineering effort is causing the problem. Migrating to liquid clustering restores full automatic maintenance.

> A compliance investigation asks for a complete row-level history of a transactions table covering the past two years. Change data feed was enabled on the table fourteen months ago and has never been disabled.
>
> The table is managed with predictive optimization enabled and default retention.

What can actually be provided?

A Fourteen months of row-level changes, from the point CDF was enabled.

B Only changes within the retention window, since CDF is not a permanent record and vacuum deletes change data files.

C Two years, since the transaction log retains all history regardless of CDF.

D Nothing, since CDF cannot be queried retrospectively.

◆ PAUSE AND DECIDE

What can actually be provided?

A Fourteen months of row-level changes, from the point CDF was enabled.

Gets the enablement rule right and forgets that retention also applies.

B Only changes within the retention window, since CDF is not a permanent record and vacuum deletes change data files.

Two limits stack: nothing before enablement, and nothing beyond retention.

C Two years, since the transaction log retains all history regardless of CDF.

Nothing before enablement was ever recorded as a change feed.

D Nothing, since CDF cannot be queried retrospectively.

It can, within its window.

ON THE EXAM

A is the near miss and the more instructive wrong answer. Half the rule is easier to remember than both halves.

A platform team migrated a set of batch jobs from classic job compute to serverless. The Databricks bill shows DBU consumption up 60%. A manager concludes the migration should be reversed.

Runtimes and schedules are unchanged.

What should be checked first?

A Nothing; a 60% rise in DBUs is a 60% rise in cost.

B The cloud provider invoice, since serverless DBU rates already include the VM cost billed separately for classic.

C Whether Photon was disabled during the migration.

D Whether autoscaling limits were carried over correctly.

◆ PAUSE AND DECIDE

What should be checked first?

A Nothing; a 60% rise in DBUs is a 60% rise in cost.
DBUs are one component, and for serverless they carry a cost that used to appear on another invoice.

B The cloud provider invoice, since serverless DBU rates already include the VM cost billed separately for classic.
Classic bills DBUs plus VMs. Serverless folds the VM into the DBU, so DBUs up and infrastructure down can be a wash.

C Whether Photon was disabled during the migration.
Would affect runtime, and the scenario says runtimes are unchanged.

D Whether autoscaling limits were carried over correctly.
Serverless does not use your autoscaling configuration.

ON THE EXAM

Total cost is DBUs plus infrastructure for classic, and DBUs alone for serverless. Judging from one invoice is the error.

> A platform team stores twelve database credentials in a single secret scope. A new requirement says the reporting team may access only the two reporting credentials.
>
> An engineer proposes granting the reporting team READ on those two secrets.

What should they be told?

A READ must be granted at scope level, so split the credentials into separate scopes.

B Grant WRITE instead, which is narrower than READ.

C Use a row filter on the secret scope to restrict which secrets are visible.

D Grant READ on the scope and rely on the reporting team not reading the others.

◆ PAUSE AND DECIDE

What should they be told?

A READ must be granted at scope level, so split the credentials into separate scopes.

Secret access control applies to the scope, not to individual secrets. Two audiences means two scopes.

B Grant WRITE instead, which is narrower than READ.

WRITE contains READ. The levels are nested, not alternatives.

C Use a row filter on the secret scope to restrict which secrets are visible.

Row filters govern table data. A secret scope is a workspace object.

D Grant READ on the scope and rely on the reporting team not reading the others.

Not a control. READ on the scope permits listing and reading every secret in it.

ON THE EXAM

C is the two-systems confusion: reaching for a Unity Catalog feature to govern a workspace object.

> A team must publish a dataset of customer purchase patterns to an external research partner. The dataset must fall outside the scope of privacy regulation entirely. Customer identifiers will be replaced with values generated by a secure tokenisation service, with the vault retained internally.
>
> Legal asks whether this meets the requirement.

Does it?

A. Yes. Tokenisation removes the identifiers, so the data is anonymised.

B. No. Tokenisation is pseudonymisation and is reversible, so the result is still personal data. Anonymisation requires suppression or generalisation.

C. Yes, provided the vault is stored in a separate catalog.

D. No, because tokenisation is not supported on Databricks.

◆ PAUSE AND DECIDE

Does it?

A Yes. Tokenisation removes the identifiers, so the data is anonymised.

Tokenisation replaces identifiers reversibly. Reversible means still personal data.

B No. Tokenisation is pseudonymisation and is reversible, so the result is still personal data. Anonymisation requires suppression or generalisation.

The requirement is to be out of scope entirely, and only anonymisation achieves that.

C Yes, provided the vault is stored in a separate catalog.

Where the vault lives does not change that re-identification remains possible.

D No, because tokenisation is not supported on Databricks.

Right conclusion, wrong reason. Tokenisation is a perfectly available technique.

ON THE EXAM

D reaches the correct verdict through a false claim, which is a distractor pattern worth recognising.

> An organisation wants every employee to be able to discover which datasets exist and request access, while ensuring nobody can read data they have not been granted.
>
> A proposal grants USE CATALOG on all catalogs to all account users, reasoning that usage privileges alone confer no data access.

What is the better approach?

A The proposal is sound; USE CATALOG alone grants no data access.

B Grant BROWSE on the catalogs, which is purpose-built for discovery and needs no usage privileges.

C Grant SELECT on all catalogs and rely on row filters.

D Grant READ METADATA on each table individually.

◆ PAUSE AND DECIDE

What is the better approach?

A The proposal is sound; USE CATALOG alone grants no data access.

The reasoning is correct and BROWSE is the purpose-built privilege, granting discovery and the ability to request access.

B Grant BROWSE on the catalogs, which is purpose-built for discovery and needs no usage privileges.

Exactly what BROWSE is for, and Databricks recommends granting it on catalogs to all account users.

C Grant SELECT on all catalogs and rely on row filters.

Grants SELECT on future tables too, so one missing filter exposes a table.

D Grant READ METADATA on each table individually.

Would work per object. BROWSE is granted once at catalog level.

ON THE EXAM

A is not wrong about the mechanics. It is just not the purpose-built answer, and it does not enable access requests.

> A governance programme wants every catalog, schema and table in the estate documented. A team proposes using AI-generated comments across the whole estate to avoid months of manual writing.
>
> There are 40 catalogs, 300 schemas and 12,000 tables.

What limitation should they plan around?

A AI-generated comments are available for tables and table columns only, not catalogs or schemas.

B AI-generated comments cannot be edited once applied.

C Comments are limited to plain text and cannot use Markdown.

D Only the metastore admin can add comments to any object.

◆ PAUSE AND DECIDE

What limitation should they plan around?

A AI-generated comments are available for tables and table columns only, not catalogs or schemas.

The documented scope. Tables and columns are the bulk of the work, and catalogs and schemas need writing by hand.

B AI-generated comments cannot be edited once applied.

They are suggestions you review and apply, and remain editable.

C Comments are limited to plain text and cannot use Markdown.

Markdown is supported in Catalog Explorer, though it does not render in DESCRIBE output.

D Only the metastore admin can add comments to any object.

The object owner or a MANAGE holder can.

ON THE EXAM

The good news is the limitation falls where the volume does not. 340 objects by hand is achievable; 12,000 is not.

> A job's task B failed. Task C depends on B and was skipped. Task D depends on A only, and completed successfully using an intermediate table that task B also writes to.
>
> An engineer repairs the run, selecting only task B.

What should they also consider?

A Nothing; repairing B automatically re-runs everything downstream of it.

B Task D succeeded against data B had partially written, so the option to re-run dependent tasks may be needed.

C Task D must be repaired first, since it completed before B.

D The whole job must be restarted, since repair cannot handle partial writes.

◆ PAUSE AND DECIDE

What should they also consider?

A Nothing; repairing B automatically re-runs everything downstream of it.

C was skipped and will re-run. D succeeded and is not a declared dependent of B.

B Task D succeeded against data B had partially written, so the option to re-run dependent tasks may be needed.

Succeeding is not the same as still being correct, and D's declared dependency hides its real one.

C Task D must be repaired first, since it completed before B.

Repair order is not something you control this way.

D The whole job must be restarted, since repair cannot handle partial writes.

Repair handles this without repeating the tasks that are still valid.

ON THE EXAM

D depends on A in the job graph and on B in reality. The graph does not know about the shared table.

> A team wants a point-in-time snapshot of a materialized view before a major pipeline change, so they can compare results afterwards. They attempt a shallow clone and it fails.
>
> They ask whether deep clone would work instead.

What should they be told?

A Deep clone works; only shallow clone is unsupported for materialized views.

B Neither works. Materialized views are unsupported as a CLONE source or target. Use CREATE TABLE AS SELECT.

C Shallow clone requires DBR 13.3 LTS, which is the actual blocker.

D Clone the underlying Delta table the materialized view is built on.

◆ PAUSE AND DECIDE

What should they be told?

A Deep clone works; only shallow clone is unsupported for materialized views.

Neither kind is supported. The restriction is on the object type, not the clone type.

B Neither works. Materialized views are unsupported as a CLONE source or target. Use CREATE TABLE AS SELECT.

A stated limitation, and CTAS produces exactly the comparison snapshot they wanted.

C Shallow clone requires DBR 13.3 LTS, which is the actual blocker.

A real requirement for shallow clones of managed tables, and not what blocks this.

D Clone the underlying Delta table the materialized view is built on.

Snapshots the input rather than the computed result, which is not the comparison they want.

ON THE EXAM

C names a genuine version requirement that is not the blocker, which is a distractor pattern worth recognising.

How to read your result

- **17 or more.** Comfortable. Compare against exam 1 and look at whether the same sections are weak.
- **13 to 16.** On track. A section you missed in both exams is worth a re-watch, not a re-read.
- **Below 13.** Work through exam 1's misses again before attempting exam 3.
- Compare the two exams by **section**, not by total. A steady total can hide one section going badly twice.

> ON THE EXAM
>
> Exam 3 is the last one, and it closes with a read of what the current guide emphasises. Save it until the other two are solid.